

9.5 The coefficient of determination and its limits

The coefficient of determination is the most widely reported summary of a regression, and probably the most widely misread. It is a ratio of two rows of the corrected table,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

defined when $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y}$ is not constant. [Section 6.7](#) showed that it is a squared cosine (Theorem 6.7.2) that cannot decrease as the model space grows. This section adds R^2 as a maximal correlation, its null distribution, the adjusted and partial versions, and what it does not measure.

9.5.1 A maximal correlation

The square root $R = \sqrt{R^2}$ is called the **multiple correlation coefficient**. The name comes from the following property, the sample counterpart of the population result Proposition 2.5.3.

Theorem 9.5.1. R is the largest correlation with a linear combination

Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{y}$ is not constant. For every $\mathbf{b}$ such that $\mathbf{Xb}$ is not constant,

$$\text{corr}(\mathbf{y}, \mathbf{Xb})^2 \leq R^2.$$

If $R^2 > 0$, equality holds iff $\mathbf{Xb} = a\hat{\mathbf{Y}} + c\mathbf{1}$ for some $a \neq 0$ and some c , and the correlation $\text{corr}(\mathbf{y}, \mathbf{Xb})$ is maximized, with value R , by the fitted values themselves. If $R^2 = 0$, every such correlation is zero.

Proof. Let $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$ and $\mathcal{V} = \mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$, whose projection is $\mathbf{M} - \mathbf{P}_1$ (Theorem 6.5.1), and let $\mathbf{v} = \mathbf{Xb} - \bar{\mathbf{Xb}}\mathbf{1}$. Since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, $\mathbf{v} \in \mathcal{V}$, so $\mathbf{u}^\top \mathbf{v} = \mathbf{u}^\top (\mathbf{M} - \mathbf{P}_1)\mathbf{v} = ((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v}$. The sample correlation is $\mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$, and by Cauchy–Schwarz (Proposition 1.1.1)

$$\text{corr}(\mathbf{y}, \mathbf{Xb})^2 = \frac{((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v})^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} \leq \frac{\|(\mathbf{M} - \mathbf{P}_1)\mathbf{u}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}},$$

with equality iff $\mathbf{v}$ and $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ are linearly dependent. If $R^2 > 0$ the second vector is nonzero, and since $\mathbf{v} \neq \mathbf{0}$ dependence means $\mathbf{v} = a(\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$ with $a \neq 0$, which is the stated condition; for $a > 0$ the correlation itself is $+R$. If $R^2 = 0$, then $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \mathbf{0}$ and $\mathbf{u}^\top \mathbf{v} = 0$. ■

Among all linear combinations of the regressors, the fitted values are the one most correlated with the response. When the regressors are pairwise uncorrelated, R^2 is the sum of the squared simple correlations (Exercise (A3)). In general it can be smaller or larger than that sum ([Exercise \(C2\)](#)).

9.5.2 R^2 when nothing is going on

Since R^2 never decreases as regressors are added, a model with many regressors has a sizeable R^2 even if none of them matters. The theory of [Chapter 4](#) says how sizeable.

Theorem 9.5.2. The distribution of R^2

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ with $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$, $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $1 < r = \text{rank}(\mathbf{X}) < n$. Let $F = [\text{SSR}/(r-1)]/[\text{SSE}/(n-r)]$ and $\gamma = \|(\mathbf{M} - \mathbf{P}_1)\boldsymbol{\mu}\|^2/\sigma^2$. Then:

- (a) $R^2 = \frac{(r-1)F}{(r-1)F + n-r}$, and $F \sim F(r-1, n-r, \gamma)$;
- (b) if $\boldsymbol{\mu} \in \text{span}(\mathbf{1})$, so that no regressor matters, then $R^2 \sim \text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$ and $E(R^2) = \frac{r-1}{n-1}$.

Proof.

- (a) $R^2/(1-R^2) = \text{SSR}/\text{SSE} = (r-1)F/(n-r)$, and solving for R^2 gives the formula. The projections $\mathbf{M} - \mathbf{P}_1$ and $\mathbf{I} - \mathbf{M}$ are orthogonal with ranks $r-1$ and $n-r$, and $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$, so by Theorem 4.4.4 $\text{SSR}/\sigma^2 \sim \chi^2(r-1, \gamma)$ and $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$, independently. The ratio F is therefore $F(r-1, n-r, \gamma)$ (Definition 4.1.2).
- (b) Now $\gamma = 0$, and $R^2 = U/(U+V)$ with $U = \text{SSR}/\sigma^2 \sim \chi^2(r-1)$ and $V = \text{SSE}/\sigma^2 \sim \chi^2(n-r)$ independent. By Exercise (B4) this is $\text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$, whose mean is $\frac{r-1}{2} / \frac{n-1}{2} = (r-1)/(n-1)$. ■

Part (b) answers the exercise Exercise (C1) of [Chapter 6](#). With $k = r-1$ regressors of pure noise, R^2 is on average $k/(n-1)$. A model with 10 useless regressors fitted to 30 observations has expected R^2 about one third. [Figure 9.5.1](#) compares the Beta densities with 100,000 simulated data sets with $n = 30$ and $k = 1, 5, 10$ regressors drawn at random. The simulated means are 0.0345, 0.1725 and 0.3445, against the exact 0.0345, 0.1724 and 0.3448. The upper 5% points of the three null distributions are 0.130, 0.353 and 0.556, so with ten regressors an R^2 of one half is unremarkable. Part (a) shows that R^2 and F carry the same information: testing R^2 against its null distribution is the overall F test of the regression.

```
import numpy as np
rng = np.random.default_rng(20260919)
n, reps = 30, 100_000
results = {}
for k in (1, 5, 10):                                     # number of regressors
    besides 1
    X = np.column_stack([np.ones(n), rng.normal(size=(n, k))])
    r = k + 1
    Q, _ = np.linalg.qr(X)
    Y = 3.0 + rng.normal(size=(n, reps))                 # no regressor matters
    Z = Q.T @ Y
    ssr = np.sum(Z[1:] ** 2, axis=0)
    sst = np.sum(Y ** 2, axis=0) - Z[0] ** 2
    R2 = ssr / sst
    R2adj = 1 - (1 - R2) * (n - 1) / (n - r)
```

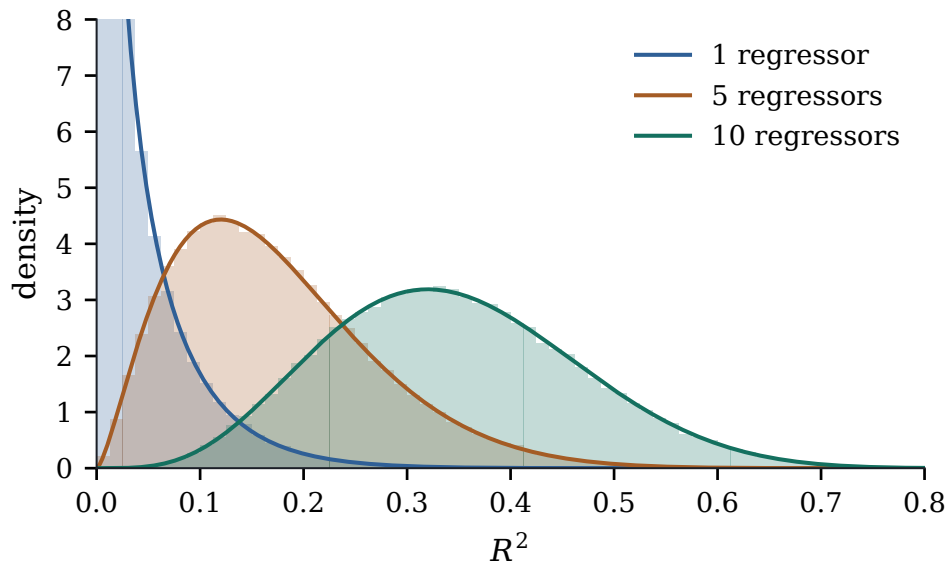


Figure 9.5.1: Null distribution of R^2 for $n = 30$ observations and 1, 5 or 10 regressors unrelated to the response. Histograms: 100,000 simulated data sets each. Curves: the Beta densities of Theorem 9.5.2.

```
print(f"k = {k:2d}: mean R^2 = {R2.mean():.4f} (theory {k / (n - 1):.4f}),"
      f" mean adjusted R^2 = {R2adj.mean():+.4f}")
results[k] = R2
```

[Run this code online](#)

9.5.3 Adjusted R^2

The bias of R^2 under the null suggests a correction. Write $s^2 = \text{SSE}/(n-r)$ and $s_y^2 = \text{SST}/(n-1)$ for the unbiased estimators of the error variance in the fitted model and in the intercept-only model. Replacing sums of squares by these mean squares defines the **adjusted coefficient of determination**

$$\bar{R}^2 = 1 - \frac{s^2}{s_y^2} = 1 - (1 - R^2) \frac{n-1}{n-r}. \quad (9.5.1)$$

Proposition 9.5.3. Properties of the adjusted R^2

Let $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, $1 \leq r = \text{rank}(\mathbf{X}) < n$, and let F be the overall F ratio of Theorem 9.5.2.

- (a) $\bar{R}^2 \leq R^2$, with equality iff $r = 1$ or $R^2 = 1$.
- (b) For $r > 1$, $\bar{R}^2 < 0$ iff $F < 1$.
- (c) If a model with rank r_0 is enlarged to one of rank $r = r_0 + q$ containing it, then $\bar{R}^2$ increases iff the F ratio of the added terms, $[\text{SS}(\text{added} \mid \text{old})/q]/s^2$, exceeds 1. For a single added column this means $|t| > 1$.
- (d) Among models for the same response, $\bar{R}^2$ is largest for the model with the smallest residual mean square s^2 .

(e) Under the normal model with $\mu \in \text{span}(\mathbf{1})$, $E(\bar{R}^2) = 0$.

Proof.

- (a) $1 - \bar{R}^2 = (1 - R^2)(n - 1)/(n - r)$, and $(n - 1)/(n - r) \geq 1$ with equality iff $r = 1$, so $1 - \bar{R}^2 \geq 1 - R^2$, with equality iff $r = 1$ or $1 - R^2 = 0$.
- (b) $\bar{R}^2 < 0$ iff $s^2 > s_y^2$, that is, $\text{SSE}(n - 1) > (\text{SSR} + \text{SSE})(n - r)$, that is, $\text{SSE}(r - 1) > \text{SSR}(n - r)$, which for $r > 1$ is $F < 1$.
- (c) s_y^2 does not depend on the model, so $\bar{R}^2$ increases iff s^2 decreases. Write $\text{SSE}_0 = \text{SSE} + D$ with D the extra sum of squares. Then $\text{SSE}/(n - r) < (\text{SSE} + D)/(n - r_0)$ iff $\text{SSE}(n - r_0) < (\text{SSE} + D)(n - r)$, iff $q \text{SSE} < D(n - r)$, iff $(D/q)/s^2 > 1$. For one column the ratio is t^2 (Corollary 9.2.3).
- (d) is immediate from Eq. 9.5.1.
- (e) By Theorem 9.5.2(b), $E(1 - R^2) = 1 - (r - 1)/(n - 1) = (n - r)/(n - 1)$, so $E(1 - \bar{R}^2) = 1$.

■

Part (e) is the point of the adjustment. In the simulation the averages of $\bar{R}^2$ are 0.0000, 0.0001 and -0.0005 , but $\bar{R}^2$ is negative in a proportion 0.674 of the data sets with one regressor: it is not a proportion of anything. Part (c) shows what selection by $\bar{R}^2$ amounts to: a regressor is admitted when $|t| > 1$, far below the threshold near two of a 5% test, so the chosen models tend to be too large (Proposition 29.2.3).

Example. Adjusted R^2 for the state data

For all seven models built from poverty, single parenthood and urbanization (each with an intercept):

Regressors	R^2	$\bar{R}^2$
poverty	0.3599	0.3466
single	0.5719	0.5629
urban	0.0221	0.0017
poverty, single	0.6410	0.6257
poverty, urban	0.4626	0.4398
single, urban	0.5844	0.5668
all three	0.6419	0.6186

R^2 is largest for the full model, as it must be. $\bar{R}^2$ prefers poverty and single parenthood without urbanization. Adding urbanization to that model raises R^2 slightly but lowers $\bar{R}^2$, because its t statistic in the full model is only 0.349, so its F ratio is $0.121 < 1$.

9.5.4 Partial R^2

Applied to what a reduced model leaves unexplained, the idea of R^2 gives a measure for a term.

Definition 9.5.1. Partial coefficient of determination

Let $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X})$ with $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$. The **partial coefficient of determination** of $\mathbf{X}$ over $\mathbf{X}_1$ is

$$R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{\text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X})}{\text{SSE}(\mathbf{X}_1)} = \frac{\text{SS}(\mathbf{X} | \mathbf{X}_1)}{\text{SSE}(\mathbf{X}_1)},$$

the proportion of the residual variation of the reduced model that the larger model explains.

Theorem 9.5.4. Partial R^2 , F and t

With the notation of [Definition 9.5.1](#), let $r_1 = \text{rank}(\mathbf{X}_1)$, $r = \text{rank}(\mathbf{X})$, $q = r - r_1 > 0$, $s^2 = \text{SSE}(\mathbf{X})/(n - r)$, and let $F = [\text{SS}(\mathbf{X} | \mathbf{X}_1)/q]/s^2$. Then:

- (a) $1 - R^2 = (1 - R_1^2)(1 - R_{\mathbf{X}|\mathbf{X}_1}^2)$, where R^2 and R_1^2 belong to the two models;
- (b) $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{qF}{qF + n - r}$;
- (c) if $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}_j]$ adds one column, then $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{t_j^2}{t_j^2 + n - r}$, and it equals the squared sample correlation between the residuals $(\mathbf{I} - \mathbf{M}_1)\mathbf{y}$ and $(\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$;
- (d) for a chain $\mathcal{C}(\mathbf{1}) = \mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1) \subseteq \cdots \subseteq \mathcal{C}(\mathbf{X}_k)$, $1 - R_k^2 = \prod_{j=1}^k (1 - R_{\mathbf{X}_j|\mathbf{X}_{j-1}}^2)$.

Proof.

- (a) $1 - R^2 = \text{SSE}(\mathbf{X})/\text{SST} = (\text{SSE}(\mathbf{X}_1)/\text{SST})(\text{SSE}(\mathbf{X})/\text{SSE}(\mathbf{X}_1))$, and the two factors are $1 - R_1^2$ and $1 - R_{\mathbf{X}|\mathbf{X}_1}^2$.
- (b) Write $D = \text{SS}(\mathbf{X} | \mathbf{X}_1) = qFs^2$ and $\text{SSE}(\mathbf{X}) = (n - r)s^2$. Then $R_{\mathbf{X}|\mathbf{X}_1}^2 = D/(D + \text{SSE}(\mathbf{X})) = qF/(qF + n - r)$.
- (c) With $q = 1$, $F = t_j^2$ by Corollary 9.2.3, and (b) gives the first formula. For the correlation, let $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{y}$ and $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$. Then $\text{SSE}(\mathbf{X}_1) = \|\tilde{\mathbf{y}}\|^2$, and by Lemma 9.2.1 $\text{SS}(\mathbf{X} | \mathbf{X}_1) = (\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}\|^2$. Their ratio is the squared cosine of the angle between $\tilde{\mathbf{y}}$ and $\tilde{\mathbf{x}}$. Both have mean zero because $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$, so the cosine is their sample correlation (Exercise (B3)).
- (d) Apply (a) repeatedly along the chain, with $R_0^2 = 0$ for the intercept-only model.

■

Part (c) identifies the partial R^2 of one regressor with a squared **sample partial correlation**, the sample counterpart of Definition 3.6.1; [Chapter 14](#) studies its sampling distribution. Within one model, ranking regressors by $|t|$ and by partial R^2 is the same thing.

Example. Partial R^2 in the state data

In the three-regressor model, with $n - r = 46$:

Regressor	t	partial R^2	R^2 without it
poverty	2.718	0.1383	0.5844

single	4.799	0.3337	0.4626
urban	0.349	0.0026	0.6410

Each partial R^2 equals $t^2/(t^2 + 46)$, and each row satisfies part (a): for poverty, $1 - 0.6419 = (1 - 0.5844)(1 - 0.1383)$. Poverty explains 14% of the variation that single parenthood and urbanization leave unexplained, although on its own it explains 36% of the total ($R^2 = 0.3599$). For the pair poverty and urbanization over single parenthood, the F ratio on 2 and 46 degrees of freedom is 4.502 and the partial R^2 is 0.1637.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
names = ["poverty", "single", "urban"]
n = len(y)

def fit(cols):
    X = np.column_stack([np.ones(n)] + [data[c].to_numpy() for c in cols])
    return sm.OLS(y, X).fit()

full = fit(names)
df_res = full.df_resid
for j, name in enumerate(names, start=1):
    reduced = fit([c for c in names if c != name])
    partial_r2 = (reduced.ssr - full.ssr) / reduced.ssr
    t = full.tvalues[j]
    print(f"{name:8s} partial R^2 = {partial_r2:.4f}, t = {t:.6.3f},"
          f" t^2/(t^2 + df) = {t ** 2 / (t ** 2 + df_res):.4f}")
print(f"R^2 = {full.rsquared:.4f}, adjusted R^2 = {full.rsquared_adj:.4f}")
```

[Run this code online](#)

9.5.5 What R^2 does not measure

A high R^2 is often read as “a good model” and a low one as “a useless model”; neither reading is safe.

It is not a property of the relationship alone. R^2 depends on the spread of the regressors, which is a property of the design. In simple regression with fixed x values, the expectations of the two sums of squares are (Exercise (B1))

$$E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}, \quad E(\text{SST}) = (n-1)\sigma^2 + \beta_1^2 S_{xx},$$

so R^2 is roughly $\beta_1^2 S_{xx} / (\beta_1^2 S_{xx} + (n-1)\sigma^2)$. In Figure 9.5.2 the line $E(y) = 2 + 0.5x$ and the same 50 errors with $\sigma = 1$ are combined with x values spread evenly over $[4, 6]$ and over $[0, 10]$. The two designs span the same $\mathcal{C}(\mathbf{X})$, so the residuals are identical and the residual mean square is 1.101 in both fits, but R^2 is 0.026 in one case and 0.632 in the other. The approximate values from the formula are 0.081 and 0.689. Restricting the range of a regressor lowers R^2 without changing the relationship.

```
import numpy as np
rng = np.random.default_rng(9)
n, beta0, beta1, sigma = 50, 2.0, 0.5, 1.0
```

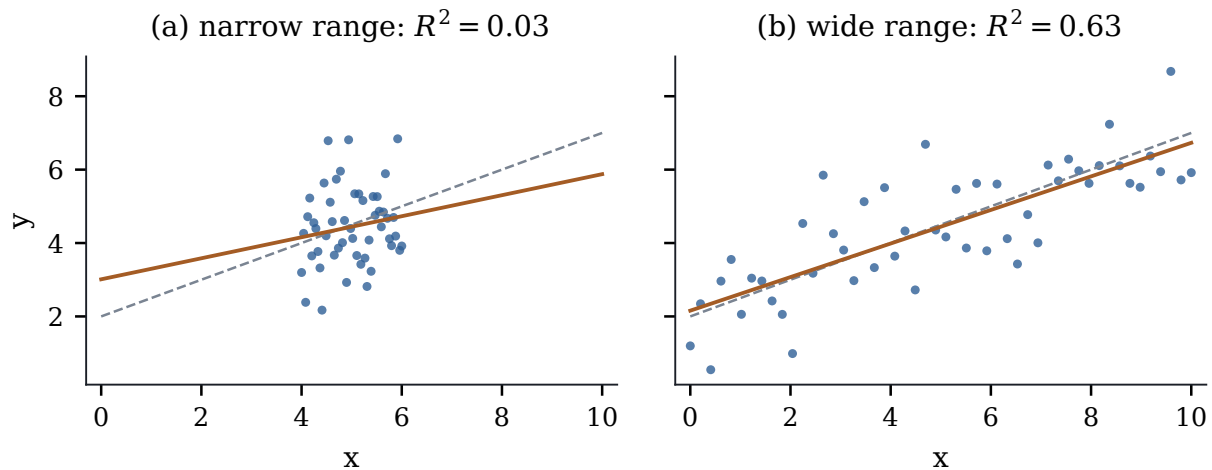


Figure 9.5.2: One line, two designs. The dashed line is the true mean $2 + 0.5x$, the solid line the least squares fit. The errors and the residual mean square are the same in both panels.

```

eps = sigma * rng.normal(size=n)                                # the same errors in both
designs                                                            designs

def fit(x):
    y = beta0 + beta1 * x + eps
    X = np.column_stack([np.ones(n), x])
    b, *_ = np.linalg.lstsq(X, y, rcond=None)
    sse = np.sum((y - X @ b) ** 2)
    r2 = 1 - sse / np.sum((y - y.mean()) ** 2)
    return y, b, sse / (n - 2), r2

for label, x in [("narrow", np.linspace(4, 6, n)), ("wide", np.linspace(0, 10, n))]:
    y, b, s2, r2 = fit(x)
    Sxx = np.sum((x - x.mean()) ** 2)
    expected = beta1 ** 2 * Sxx / (beta1 ** 2 * Sxx + (n - 1) * sigma ** 2)
    print(f"{label:6s}: slope {b[1]:.3f}, s^2 {s2:.3f}, R^2 {r2:.3f}, design value {
    expected:.3f}")

```

[Run this code online](#)

It is not a test of the model. A line fitted to points on a smooth curve can have a high R^2 while missing the curvature; residual plots and lack-of-fit tests (Theorem 14.6.2) reveal it. Conversely, a correct model has a low R^2 whenever the error variance is large, and its coefficients can still be precisely estimated and important.

It is not comparable across responses. Models for y and $\log y$, or for individuals and for group averages, have different denominators SST. Averaging replicates inflates R^2 (Exercise (B2)).

It is not comparable without an intercept (Proposition 9.1.3, Example (Forcing the murder regression through the origin)).

It is not a measure of prediction or of cause. Computed on the fitting data, it overstates predictive accuracy for new data (Chapter 29), and variation “explained” is linear association, not the effect of changing a regressor (Chapter 25).

9.5.6 Exercises

9.5.6.1 A. Check your understanding

Exercise. A1

A regression with an intercept, $n = 20$ and $r = 6$ has $R^2 = 0.40$. Compute $\bar{R}^2$ and the overall F ratio. Is $\bar{R}^2$ positive? Is it consistent with part (b) of [Proposition 9.5.3](#)?

9.5.6.2 B. Practice

Exercise. B1

In simple regression with fixed x values, $E(\mathbf{Y}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, use Theorem 2.3.1 to show that $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(\text{SST}) = (n-1)\sigma^2 + \beta_1^2 S_{xx}$. Show that the ratio of these expectations tends to 1 as $S_{xx} \rightarrow \infty$ with n fixed, and to $1/(n-1)$ as $S_{xx} \rightarrow 0$, and interpret the second limit.

Solution. $\text{SSR} = \mathbf{y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{y}$, where $\mathbf{M} - \mathbf{P}_1$ projects onto the line spanned by $\mathbf{x} - \bar{x} \mathbf{1}$ (Example (Intercept inside simple regression)) and has trace 1. With $\boldsymbol{\mu} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$, $(\mathbf{M} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$, whose squared length is $\beta_1^2 S_{xx}$. Theorem 2.3.1 gives $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$. Similarly $\text{SST} = \mathbf{y}^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{y}$, with trace $n-1$ and $(\mathbf{I} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$. The ratio $(\sigma^2 + \beta_1^2 S_{xx}) / ((n-1)\sigma^2 + \beta_1^2 S_{xx})$ tends to 1 as $S_{xx} \rightarrow \infty$ and to $1/(n-1)$ as $S_{xx} \rightarrow 0$. The second limit is the null value of $E(R^2)$ for $r = 2$ ([Theorem 9.5.2](#)): with almost no spread in x , the slope contributes almost nothing and the regression sum of squares is almost pure noise.

Exercise. B2

Suppose each of g distinct values $x_1, \dots, x_g$ carries m observations. Show that the least squares line fitted to all gm observations is the same as the line fitted to the g group means, and that its R^2 for the individual data is at most its R^2 for the means, with equality iff every observation equals its group mean.

Solution. The model space for the individual data is spanned by $\mathbf{1}$ and $\mathbf{x}$, both constant within groups, so it lies inside the space of group-constant vectors. By Theorem 6.5.1, projecting $\mathbf{y}$ onto it equals projecting the vector of group means (repeated m times) onto it, which is the fit to the means with each point weighted equally. Write SSR_g and SST_g for the sums over the g means, and W for the within-group sum of squares. For the individual data $\text{SSR} = m \text{SSR}_g$ and $\text{SST} = m \text{SST}_g + W$. Hence $R^2 = m \text{SSR}_g / (m \text{SST}_g + W) \leq \text{SSR}_g / \text{SST}_g$, with equality iff $W = 0$.

Exercise. B3

Using part (d) of [Theorem 9.5.4](#) and the sequential sums of squares of Example (Order

matters) (order A), express $1 - R^2$ for the three-regressor model as a product of three factors, and identify each factor as one minus a partial R^2 .

9.5.6.3 C. Going deeper

Exercise. C1

Under the conditions of [Theorem 9.5.2](#), show that for every $c \in (0, 1)$ the probability $\Pr(R^2 > c)$ is a strictly increasing function of the noncentrality γ . *Hint:* part (a) of the theorem and Theorem 4.1.4.

Exercise. C2

For two regressors with sample correlations $r_{y1} = 0.5$, $r_{y2} = 0$ and $r_{12} = 0.6$, compute R^2 and compare it with $r_{y1}^2 + r_{y2}^2$. Use the formulas of Exercise (B1). Explain geometrically why a regressor that is uncorrelated with the response can raise R^2 .